$$y=\sqrt{x}+8\sqrt[3]{x}-2\sqrt[4]{x}$$

$$y=x^{\frac{1}{2}}+8x^{\frac{1}{3}}-2x^{\frac{1}{4}}$$

$$\frac{dy}{dx}=\frac{1}{2}x^{-\frac{1}{2}}+\frac{8}{3}x^{-\frac{2}{3}}-\frac{1}{2}x^{-\frac{3}{4}}$$

$$\frac{dy}{dx}=\frac{1}{2\sqrt{x}}+\frac{8}{3\sqrt[3]{x^2}}-\frac{1}{2\sqrt[4]{x^3}}$$